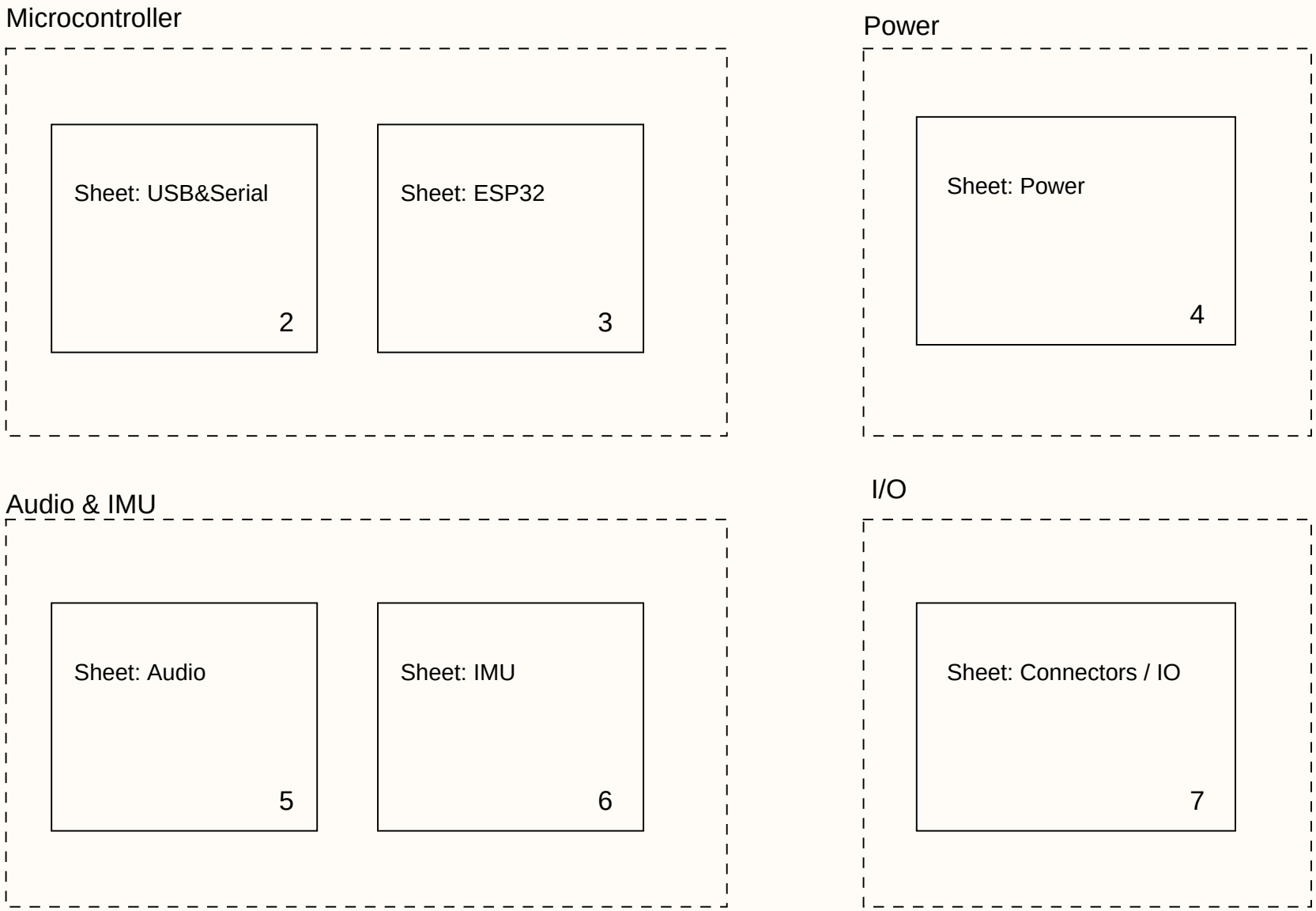
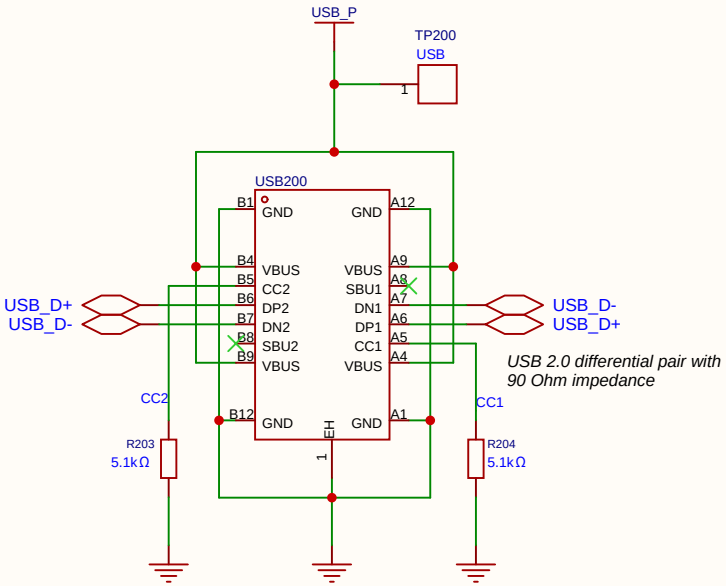


Schematic	Remote_R2	Update Date	2025-01-01
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Page	Overview	Drawn	Christoph Streit
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EasyEDA	1 / A4	Hobbyist	

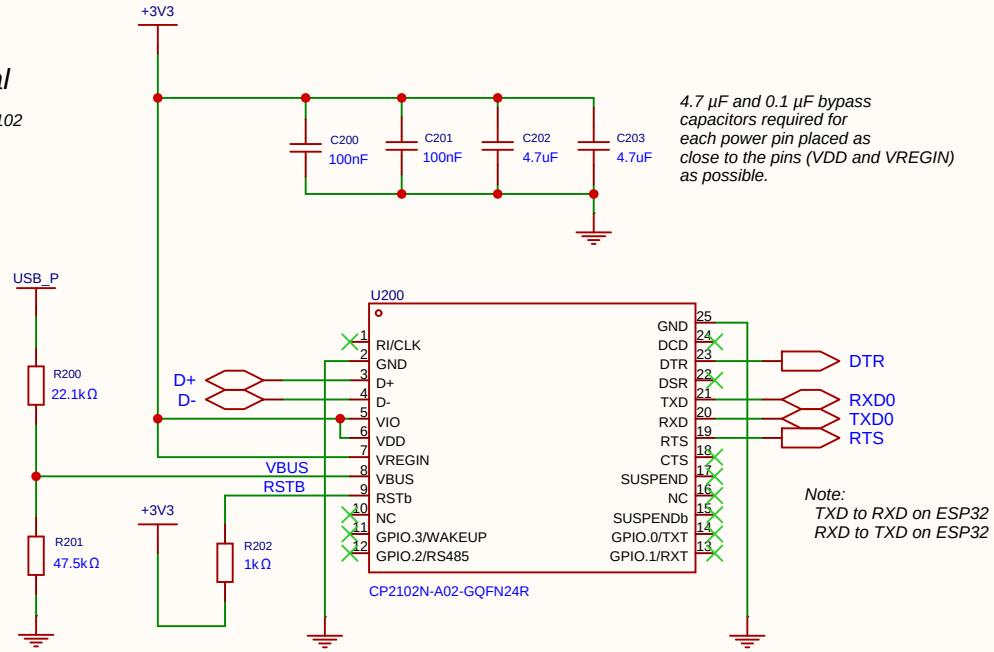
USB



USB 2.0 differential pair with 90 Ohm impedance

USB to Serial

Bus-powered configuration of the CP2102

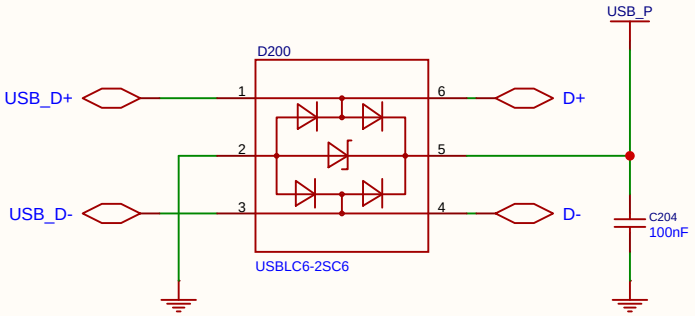


4.7 µF and 0.1 µF bypass capacitors required for each power pin placed as close to the pins (VDD and VREGIN) as possible.

Note:
TXD to RXD on ESP32
RXD to TXD on ESP32

Both SUSPEND and SUSPENDb temporarily float high during a CP2102N reset. If this behavior is undesirable, a strong pull-down (10 k\&) can be used to ensure SUSPENDb remains low during reset.
--> not installed

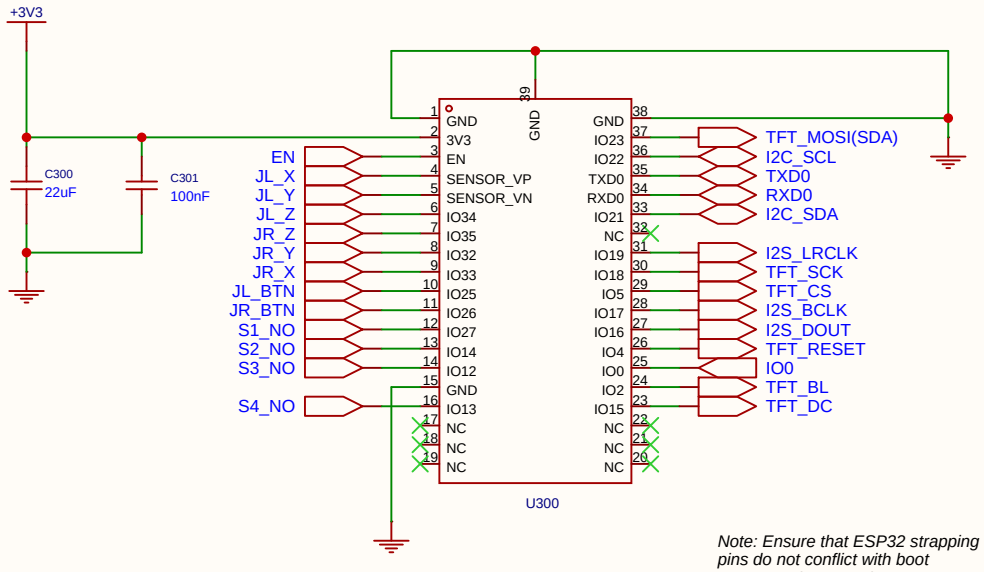
ESD Protection

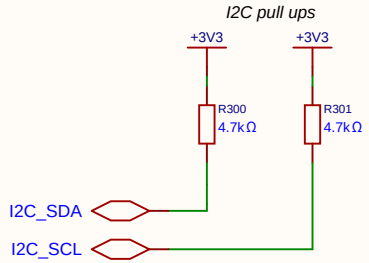


Datasheet: https://www.silabs.com/documents/public/data-sheets/cp2102n-datasheet.pdf

Schematic	Remote_R2	Update Date	2025-01-01
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Page	USB+Serial	Drawn	Christoph Streit
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EasyEDA	1	A4	Hobbyist

ESP32--WROOM-32E

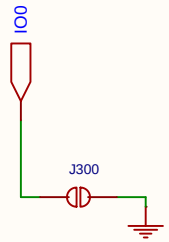




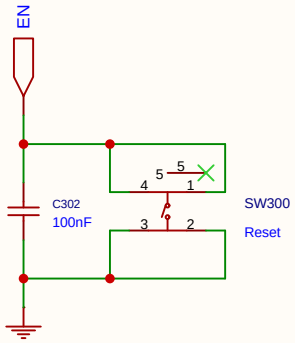
Note: Ensure that ESP32 strapping pins do not conflict with boot sequence (e.g. IO02).

Datasheet: *https://www.espressif.com/sites/default/files/documentation/esp32-wroom-32e_esp32-wroom-32ue_datasheet_en.pdf*

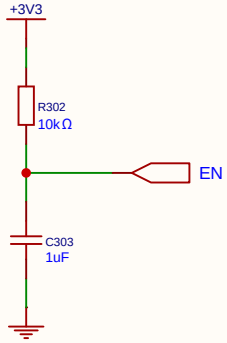
Boot Option and Reset Button



Boot Option to put the board in flashing mode, e.g. short the solder jumper with tweezers.

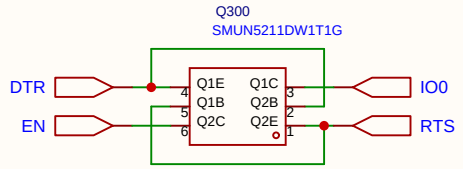


Press reset button to reset (restart) the board.



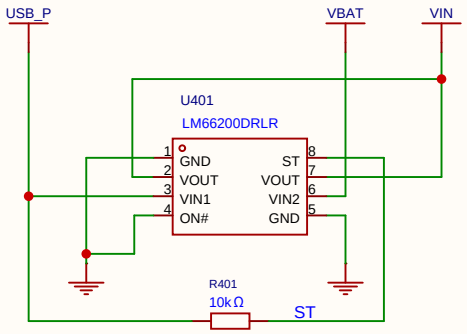
RC delay circuit ensures power supply is stable during power-up

Autoreset



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Page		ESP32		Drawn	Christoph Streit
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EasyEDA		1	A4	Hobbyist	

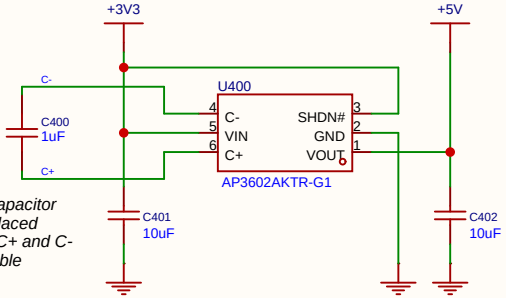
Automatic Switchover



Option: Attach ON# to 1M resistor and then to GND (see Adafruit schematics) in order to turn the device on/off by the CPU

Datasheet: https://www.ti.com/lit/ds/symlink/lm66200.pdf

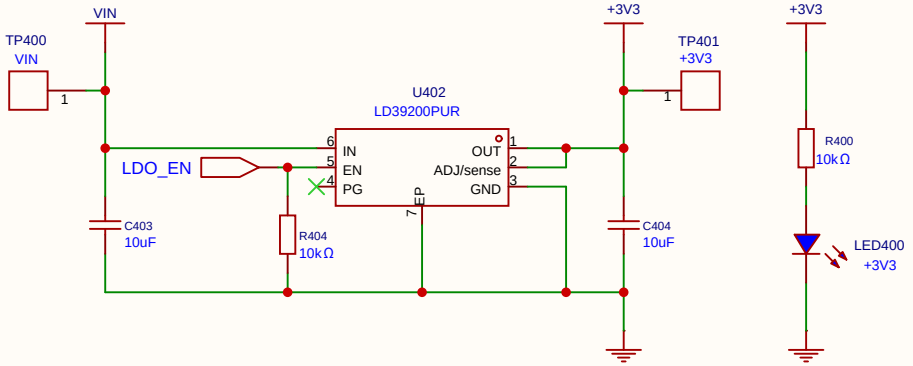
Charge Pump (+5V)



The flying capacitor should be placed as close to C+ and C- pin as possible

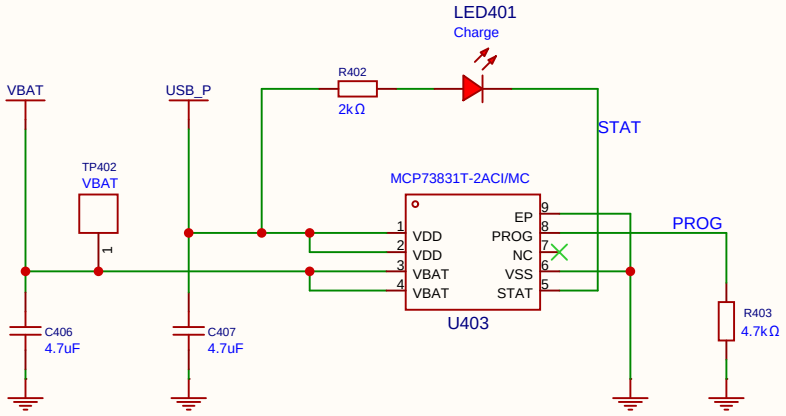
3.3V LDO

3.3V LDO to power the ESP, the CP2102 bridge, the display and other devices



Datasheet: https://www.st.com/resource/en/datasheet/ld39200.pdf

Battery Charger



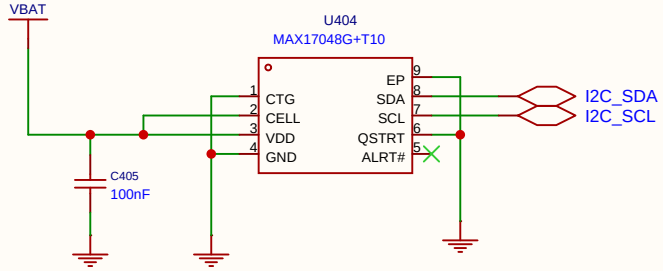
Ireg = 1000V/Rprog

500mA = 1000v/2k (max) -> hot
100mA = 1000v/10k

https://ww1.microchip.com/downloads/en/DeviceDoc/MCP73831-Family-Data-Sheet-DS20001984H.pdf

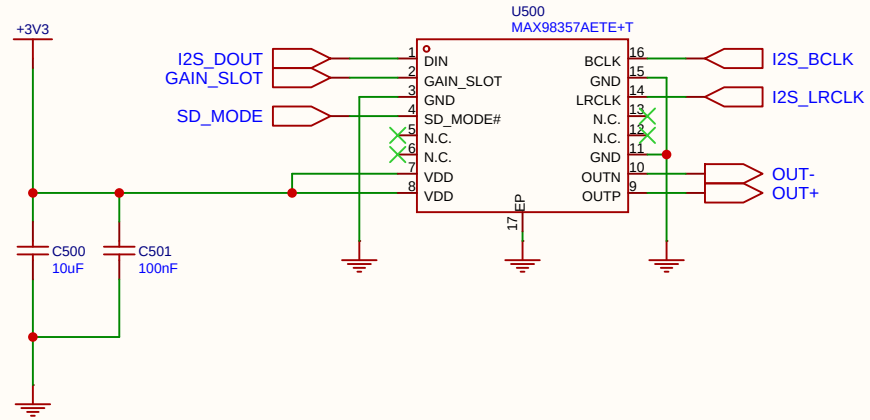
Battery Gauge

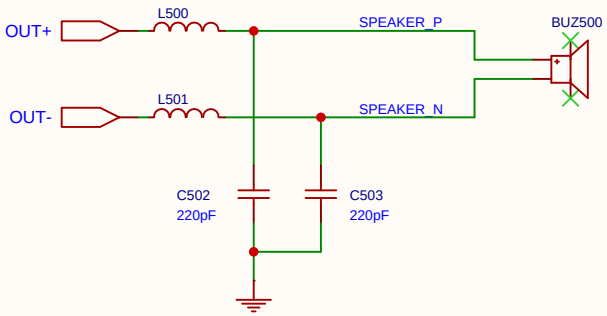
Layout hint: IN/GND with Kelvin (4-wire) setup



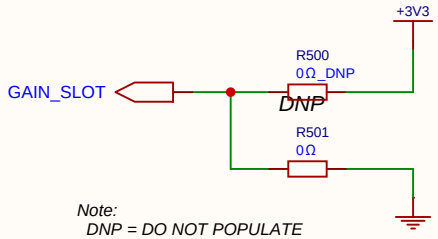
Schematic	Remote_R2		Update Date	2025-08-11
			Create Date	2025-08-11
Page		Power	Drawn	Christoph Streit
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EasyEDA		1 / A4	Hobbyist	

Amplifier





Gain Selection	
GAIN_SLOT	GAIN
Connect to GND through 100kΩ resistor	+15 dB
Connect to GND	**+12 dB**
Unconnected	+9 dB
Connect to VDD	+6 dB
Connect to VDD through 100kΩ resistor	+3 dB



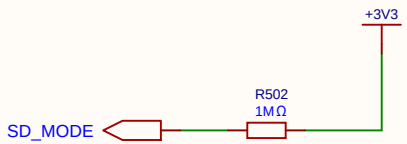
Note:
DNP = DO NOT POPULATE

Channel Selection (SD_MODE)

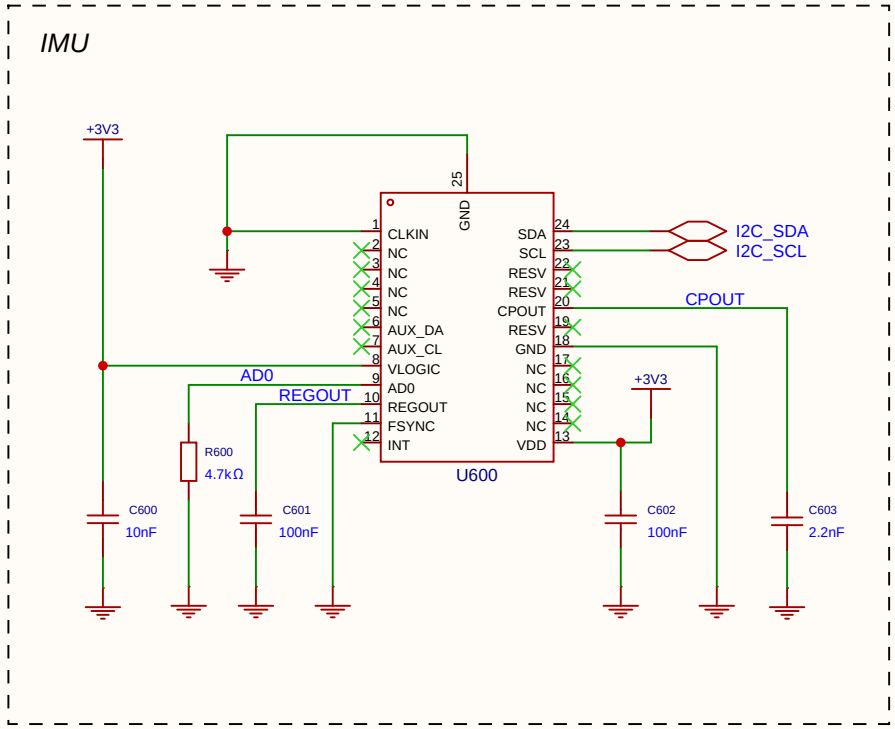
LEFT = VDD (0 Ω)

RIGHT = 370K
= 94*5-100 (RSMALL (kΩ) = 94.0 x VDDIO - 100)

LEFT/2 + RIGHT/2 = 1.011M
= 222.2*5-100 (RLARGE (kΩ) = 222.2 x VDDIO - 100)

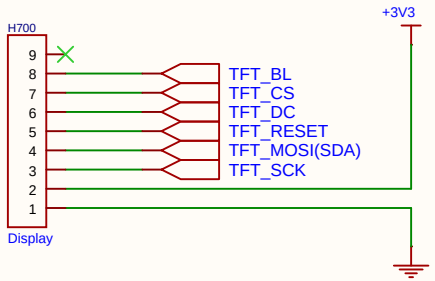


Schematic	Remote_R2		Update Date	2025-01-02
			Create Date	2025-01-02
Page		Audio	Drawn	Christoph Streit
		VER	SIZE	PAGE 5 OF 7
EasyEDA		1	A4	Hobbyist

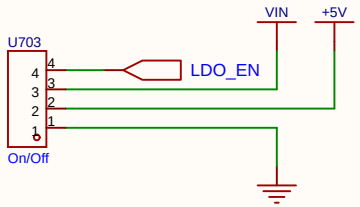


Schematic	Remote_R2			Update Date	2025-01-01
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		VER	SIZE	PAGE 6 OF 7	
EasyEDA		1	A4	Hobbyist	

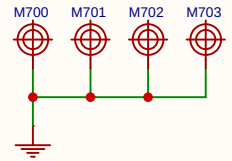
Display



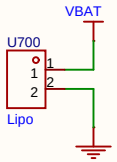
On/Off



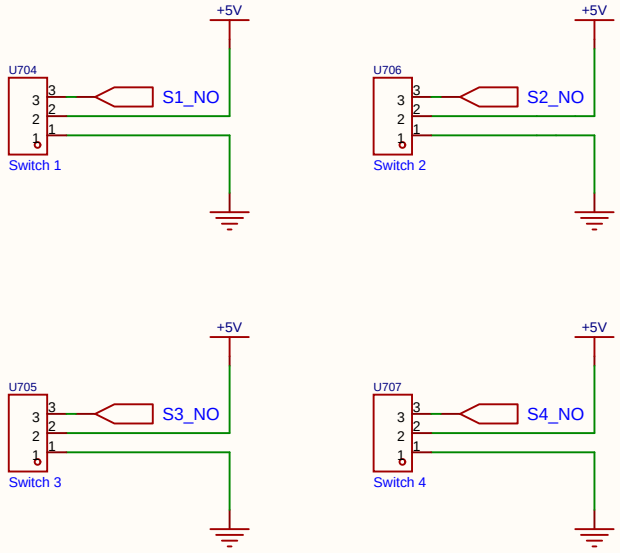
M2 Mounting Holes



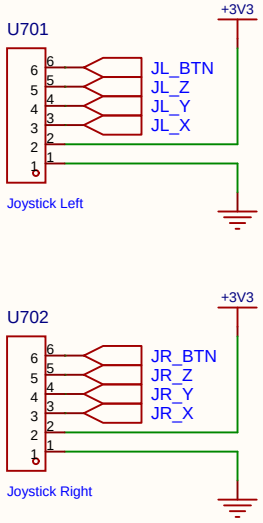
LiPo



Switches



Joysticks



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